INTERNSHIP REPORT

Visual Geolocalization of UAV images using registration technique with satellite imagery

by

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Under the guidance of
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DECLARATION

I hereby declare that the entire work embodied in this dissertation has been carried out by me and no part of it has been submitted for any degree or diploma of any institution previously.

Signature of the student

Place:
Date: Name:

CERTIFICATE

04 – 08 - 2026

This is to certify that S S S N Sameer Kumar Enrollment No 230021018 student of IIT INDORE pursuing Btech in Space Science and Engineering has successfully completed Summer Internship titled *Visual geolocalization of UAV images using registration technique with satellite imagery* during June 4 to August 4 under the guidance of Mrs Devshree Kumar, Sr Principal Scientist, UAV Division towards the fulfilment of the requirements of their University.

We wish Sameer Kumar the very best in all his future endeavours.

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S S S N Sameer Kumar

1. Abstract

Visual geolocalization estimates the ground position of an airborne camera or a specific target within its field of view by matching live drone imagery against a georeferenced satellite base map. This capability serves as an essential, unjammable navigation fallback when Global Navigation Satellite System (GNSS) signals are lost, jammed, or spoofed. Developed as an engineering internship project, this report details a modular, Python-driven pipeline built to execute automated target geolocalization from UAV frames onto satellite orthomosaics. The primary technical hurdle is bridging the severe visual and geometric domain gap between low-altitude, oblique drone views and top-down satellite imagery, which differ drastically in spatial resolution, lighting, viewing angles, and camera lens dynamics. To resolve these disparities, the system executes a sequential multi-stage workflow: initial UAV frame orientation alignment using onboard IMU telemetry and CLAHE contrast equalization to balance lighting; detector-free dense feature matching via a Local Feature Transformer (LoFTR), which replaces fragile traditional keypoint detectors like SIFT or ORB; robust geometric filtering using MAGSAC++ to eliminate false matches and compute a planar homography; perspective warping to overlay the drone footprint onto the satellite canvas; and spatial translation of selected target pixels into WGS84 geographic coordinates using embedded GeoTIFF metadata or a four-corner calibration fallback. Opting for a deep transformer-based matcher over classical feature detectors prioritizes feature-matching resilience across large visual gaps over raw computational speed. Experimental testing confirms that combining LoFTR with MAGSAC++ reliably isolates true ground points and generates geometrically consistent correspondence fields. However, the system's reliance on a single planar homography assumption introduces horizontal position drift over steep terrain or tall vertical structures. The report concludes by evaluating these geometric boundaries and outlining concrete operational upgrades, including Digital Elevation Model (DEM) integration, real-time sensor fusion, and model optimization for deployment on resource-constrained flight controllers.

2. Introduction

2.1 Operational Context: Navigating in GNSS-Denied Airspace

Modern Unmanned Aerial Vehicles (UAVs) rely almost universally on onboard Inertial Measurement Units (IMUs) tied to Global Navigation Satellite System (GNSS) receivers to keep their bearings and track ground targets. In practice, however, satellite links are fragile. They are easily disrupted by urban signal blocking, multipath reflections, active RF jamming, or malicious GPS spoofing.

Visual geolocalization offers a self-contained, unjammable navigation alternative. By matching real-time optical frames from an onboard drone camera against a georeferenced satellite base map, the aircraft can independently work out its global position—or calculate the exact coordinates of ground targets—without relying on external satellite signals.

2.2 Domain Gap and Technical Challenges

Cross-view matching between low-altitude drone images and satellite reference maps involves bridging a massive visual and structural domain gap. In real-world environments, this difficulty is compounded by several factors:

- Severe Viewpoint Disparity: Satellite basemaps are strictly top-down (nadir) orthophotos. Drone cameras, by contrast, frequently shoot at low altitudes and oblique angles, introducing significant perspective distortion, foreshortening, and visible building facades.

- Radiometric and Sensor Variation: Differences in hardware sensors, flight altitudes, sun angles, seasonal foliage, and Ground Sample Distance (GSD) mean that the same patch of ground can look drastically different across the two image sources.

Traditional registration techniques relied on handcrafted feature detectors to spot key interest points. While these work fine on similar photos taken from identical angles, they break down when faced with extreme perspective gaps. Many older systems also assume the ground is completely flat or demand complex 3D surface models, making them unreliable when flying uncalibrated missions over uneven terrain.

2.3 System Formulation & Coordinate Mapping

The core task of our pipeline is straightforward: take a target pixel coordinate selected from a UAV video frame and map it to its true geographic location (latitude and longitude in the WGS84 system) on a reference satellite basemap.

We break this spatial translation down into a two-stage process:

1. Perspective Alignment: We compute a projective transformation (planar homography) that warps the perspective of the drone frame to fit the coordinate layout of the reference satellite map. This bridges the physical camera angle differences and aligns matching pixels across both views.

2. Geographic Translation: Once the target pixel is mapped onto the satellite image coordinate space, a secondary translation converts those image pixels into real-world geographic coordinates. This is resolved using embedded spatial metadata within the map file or calibrated via known corner positions.

Every module in the system exists to keep this mapping accurate: raw frames are equalized to balance lighting, dense feature matching connects shared landmarks across platforms, iterative consensus algorithms filter out false matches, and the georeferencing engine converts the final pixel locations into ground coordinates.

2.4 Implementation Scope & System Modules

To test and validate this workflow, we implemented a modular Python framework evaluated on the benchmark UAV-VisLoc dataset, which provides paired UAV imagery, telemetry, and high-resolution satellite orthomomsaics. The software pipeline is divided into six core functional components:

- Main Execution Control: A single pipeline manager (main.py) that ingests UAV and satellite pairs from the dataset, runs the processing steps end-to-end, and outputs visual diagnostics alongside the target ground location.

- Radiometric Conditioning: A contrast enhancement module using adaptive histogram equalization (CLAHE) to normalize brightness and exposure differences between the two camera sensors.

- Dense Transformer Matching: A detector-free matching engine powered by a Local Feature Transformer (LoFTR) that establishes dense, coarse-to-fine correspondences across wide viewpoint gaps without fragile keypoint detection.

- Robust Outlier Rejection: A geometric solver using MAGSAC++ to scrub bad feature matches and calculate an optimal homography matrix.

- Footprint Overlay Verification: A visualization module that warps the UAV image frame into the satellite image geometry, rendering an alpha-blended footprint overlay to visually confirm alignment quality.

- Dual-Mode Georeferencing Utility: A spatial translation script that automatically extracts raster metadata via rasterio to calculate latitude and longitude, with a four-corner perspective mapping fallback when spatial tags are missing.

3. Literature Review & Theoretical Foundations

3.1 Feature Detection Paradigms in Remote Sensing

For decades, cross-sensor spatial alignment relied heavily on handcrafted keypoint detectors and hand-engineered local descriptors. Among these, the Scale-Invariant Feature Transform (SIFT) detects stable interest points across scale-space pyramids built using Difference-of-Gaussians operators, representing each landmark through a 128-dimensional vector of local gradient orientations. Speeded-Up Robust Features (SURF) accelerates this workflow by applying box filters over integral images, though it trades off some descriptor uniqueness. Oriented FAST and Rotated BRIEF (ORB) prioritizes raw computational speed by extracting binary bitstrings around FAST corner locations, enabling high-speed matching using Hamming distance logic.

Despite their success in same-sensor conditions, these classical frameworks perform poorly when bridging the aerial-to-satellite domain gap. Multi-platform observations introduce sharp differences in camera perspectives, ground sample distances, and sun angles. These variations break the repeatability of Gaussian extrema detectors and severely degrade the distinctiveness of gradient-based descriptors.

To handle these structural gaps, modern visual geolocalization architectures increasingly turn to multi-stage deep learning pipelines. For instance, Qiu et al. (2025) introduced a hierarchical framework that queries an offline image database using coarse global features (combining VGG16, NetVLAD, and PCA) to pull candidate satellite patches. Fine alignment is subsequently handled by a detector-free keypoint network incorporating SuperPoint and SuperGlue. To reconcile scale differences across views, their system applies a Spatial Resolution Adaptation module before solving for full six-degree-of-freedom platform poses using Perspective-n-Point algorithms integrated with digital elevation models.

Alongside global feature retrieval, spatial filtering via sliding window search offers another effective strategy for isolating ground regions. By systematically scanning across large, high-resolution satellite GeoTIFF files using segmented windowing operations, this method processes spatial tiles incrementally. Keeping memory consumption bounded to a single tile prevents system crashes while evaluating keypoint densities across

candidate areas. As a result, high-probability regions of interest can be isolated before executing compute-heavy fine registration steps.

While sliding-window filtering provides a practical search strategy, state-of-the-art systems frequently adopt the Local Feature Transformer (LoFTR) for fine matching. By bypassing keypoint detection entirely, LoFTR produces dense representations that overcome the sparse keypoint failures common to classical detectors. Classical algorithms and global retrieval systems remain foundational benchmarks for measuring the performance of these dense deep-learning strategies.

3.2 Image Preprocessing & Contrast Normalization

Standard global histogram equalization stretches an image's intensity range across a uniform distribution. While this boosts overall contrast, it frequently amplifies background noise in flat areas and washes out subtle textures in regions marked by strong lighting variations—a common issue in raw aerial and satellite photos.

Contrast-Limited Adaptive Histogram Equalization (CLAHE) overcomes these issues by dividing the image into a grid of small, non-overlapping contextual tiles. Each local tile undergoes independent equalization, with histogram values capped at a specified clip limit to prevent noise amplification before contrast redistributions occur. Smooth transitions across tile borders are then restored using bilinear interpolation.

This localized approach provides far better adaptivity against uneven sunlight and deep shadows. Combining adaptive contrast adjustments (using OpenCV's CLAHE routines) with light Gaussian filtering forms a dependable preprocessing pipeline, smoothing out high-frequency noise created during contrast enhancement before downstream feature extraction begins.

3.3 Geometrical Ray-Casting & Spatial Photogrammetry

Rigorous photogrammetric mapping projects image coordinates into geographic space using the camera's internal parameters (focal length and principal point alignment) alongside its external orientation angles (pitch, yaw, roll, and spatial altitude) relative to a global coordinate system. A target pixel projects outward as a 3D ray in camera space, which is transformed into world coordinates via orientation matrices and intersected against a digital terrain model or flat elevation plane. The resulting ground point is then converted into geographic latitude and longitude through standard coordinate transformations. Advanced systems, such as the one proposed by Qiu et al. (2025), combine 3D terrain models with Perspective-n-Point solvers to calculate spatial position parameters directly.

When precise internal lens calibrations or exterior orientation angles are unavailable due to missing telemetry, direct 3D ray-casting becomes unfeasible. Under these constraints, a 2D projective transformation—known as a planar homography—serves as a practical mathematical proxy, mapping the aerial coordinate system straight onto the reference satellite plane. This model relies on the assumption that the underlying terrain is relatively flat, allowing the spatial transformation to be expressed as a single transformation matrix.

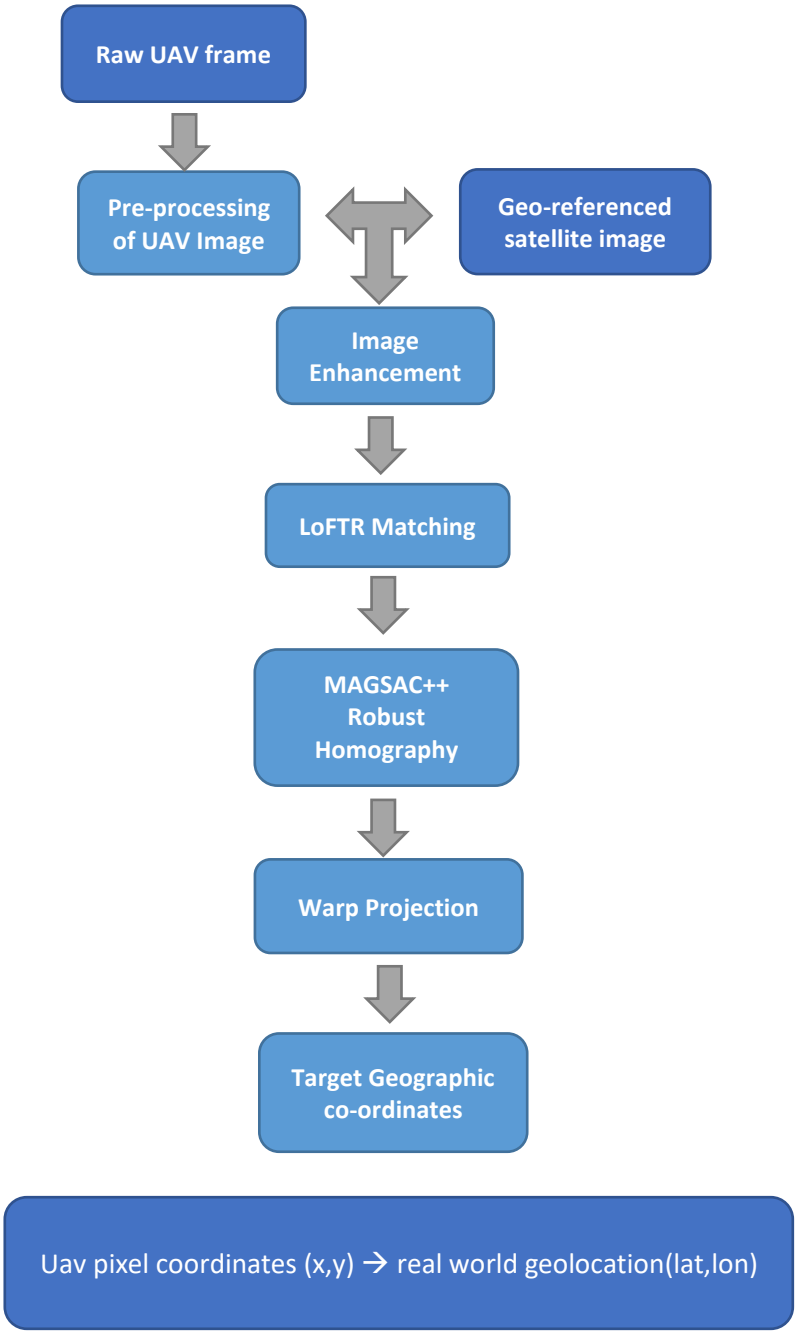
Conversely, converting satellite pixel coordinates into real-world geographic coordinates remains exact as long as valid GeoTIFF spatial tags are present. Parsing native affine transformation matrices and coordinate reference systems allows direct, mathematically rigorous reprojection into standard global mapping systems such as EPSG:4326 (WGS84).

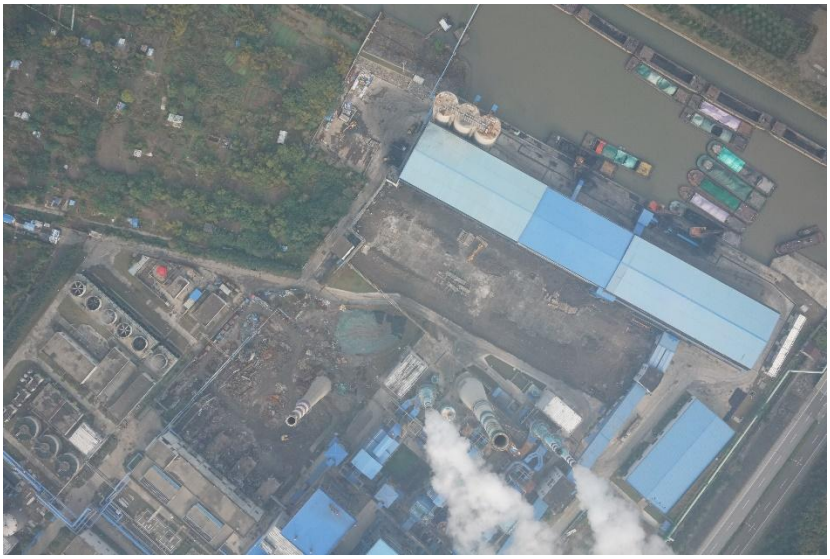
4. Methodology & Implementation

4.1 Overall Architecture Overview

The system operates as an end-to-end processing pipeline that translates target pixel coordinates from an uncalibrated UAV frame into WGS84 geographic coordinates on a satellite basemap. The workflow begins with telemetry-guided orthorectification and heading alignment, followed by adaptive contrast enhancement (CLAHE) to normalize lighting. A detector-free Local Feature Transformer (LoFTR) extracts dense feature correspondences across both views, which are geometrically filtered using MAGSAC++ to compute a planar homography matrix. Finally, the UAV frame is warped onto the satellite grid for visual verification, and target pixels are converted into geodetic coordinates via embedded GeoTIFF metadata or a four-corner calibration fallback

WORKFLOW


Raw UAV frame
Pre-processing of UAV Image
Geo-referenced satellite image
Image Enhancement
LoFTR Matching
MAGSAC++ Robust Homography
Warp Projection
Target Geographic co-ordinates
Uav pixel coordinates (x,y) → real world geolocation(lat,lon)



Uav_Image(raw)



Satellite_iamge(georeferenced)

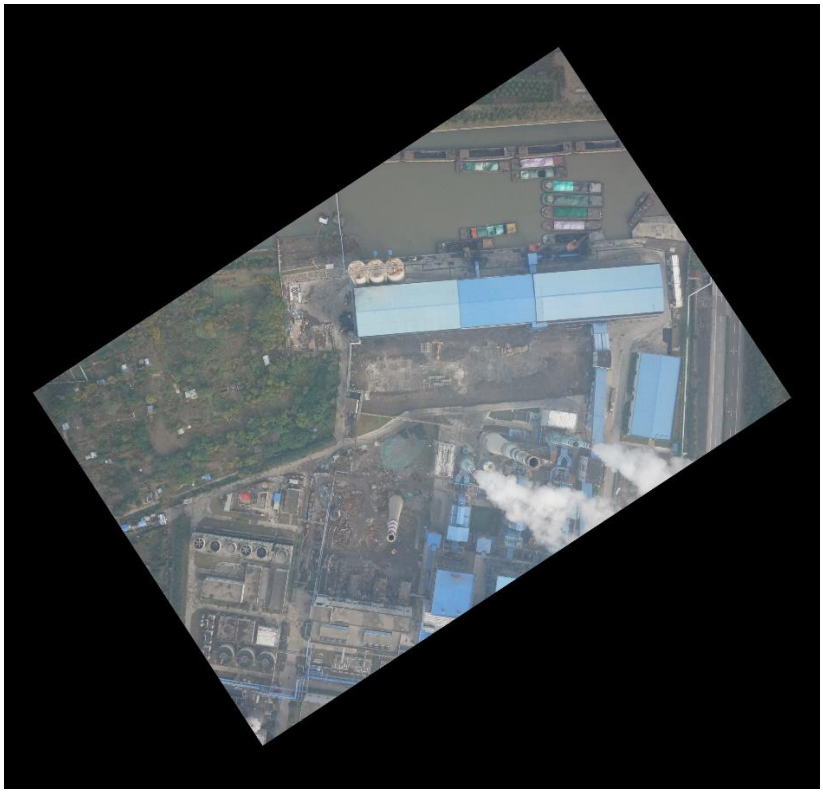
4.2 Step 0: UAV Image Pre-processing

Before the image pair enters the main feature-matching stage, the raw aerial image undergoes two sequential geometric transformations to establish an initial spatial pre-alignment:

- Perspective Orthorectification via Telemetry: The system uses flight attitude parameters recorded by the aircraft's onboard Inertial Measurement Unit—specifically pitch, roll, and yaw angles—alongside the drone's flying altitude. This step removes perspective skew, angled distortions, and foreshortening caused by camera tilt and elevation differences. By projecting the raw, angled drone capture onto a synthetic ground grid, it converts the oblique perspective into an orthorectified top-down representation that mimics an overhead camera view.

- Heading Rotation Alignment: After fixing the perspective, the drone frame is rotated to match the standard North-up geographic orientation of the reference satellite map. A two-dimensional rotation is computed around the image center based on the heading offset between the drone platform and the reference map. This rotation is applied using an expanded canvas size to prevent any corners of the image from being cut off during transformation.

  Pre-aligning both the camera perspective and heading direction drastically reduces the visual and structural gap between the two image sources. This ensures the downstream deep transformer matcher can easily identify shared landmarks and establish accurate, reliable feature correspondences.



4.2 Step 1: Image Enhancement

The initial conditioning stage subjects both the aerial frame and the reference satellite scene to three sequential radiometric processing steps:

- Grayscale Conversion**:** Full-color inputs are first converted into single-channel intensity images. This removes color channel variations that downstream feature-matching modules do not use, focusing spatial alignment entirely on structural shapes and intensity patterns. Dropping color channels is a practical necessity because color response varies widely between standard visual cameras on drones and multispectral or processed satellite images.
- Local Contrast Enhancement**:** Contrast-Limited Adaptive Histogram Equalization (CLAHE) is applied to the grayscale frames to boost local contrast. This brings out fine surface textures in shadowed or overly bright areas that would otherwise remain hidden.
- Noise Suppression Filtering**:** A light spatial Gaussian filter using a minimal smoothing window is applied to the contrast-boosted images. This step dampens high-frequency sensor grain and noise that localized contrast boosting naturally tends to amplify. Keeping the smoothing window small ensures noise is filtered out without blurring sharp structural edges or object boundaries needed for feature alignment.

A side-by-side visual comparison of both processed grayscale frames serves as a quality control checkpoint. This step allows engineers to inspect local contrast quality and verify that structural boundaries remain crisp before passing the images into the more compute-heavy feature-matching pipeline.

Parameter	Value / Definition (config.py)	Purpose
CLAHE_CLIP_LIMIT	2.0	Caps the local histogram bin height before redistribution; a moderate value that boosts local contrast without over-amplifying sensor noise in near-uniform regions (water, bare soil, rooftops).
CLAHE_TILE_GRID_SIZE	(8, 8)	Number of tiles the image is divided into for local equalization; 8×8 balances adaptivity to local illumination against blocking artifacts and computational cost.
Gaussian kernel	(3, 3), σ=0	Light smoothing post-CLAHE to suppress amplified high-frequency noise prior to feature matching;

Parameter	Value / Definition (config.py)	Purpose
		σ=0 lets OpenCV derive sigma automatically from the kernel size.

Table 4.1 — Step 1 enhancement parameters (config.py).



4.3 Steps 2–3: Detector-Free Deep Feature Matching

Instead of relying on classical detect-describe-match algorithms like SIFT or ORB, the system uses a detector-free matching framework powered by a Local Feature Transformer (LoFTR). Rather than searching for isolated corner points first, LoFTR extracts dense, coarse-level feature maps directly using a convolutional neural network. It then refines these features using alternating self-attention (within the same frame) and cross-attention (between the UAV and satellite frames) transformer layers. By evaluating global contextual relationships across full feature maps, the network handles severe differences in perspective, scale, and lighting far better than traditional point-matching tools.

The feature matching workflow executes through four main phases:

1. Image Rescaling and Grid Alignment: Input grayscale images are capped at a maximum edge length (such as 840 pixels) to keep memory usage under control. Both width and height dimensions are rounded down to the nearest multiple of eight so they align cleanly with the feature-map stride of the underlying network

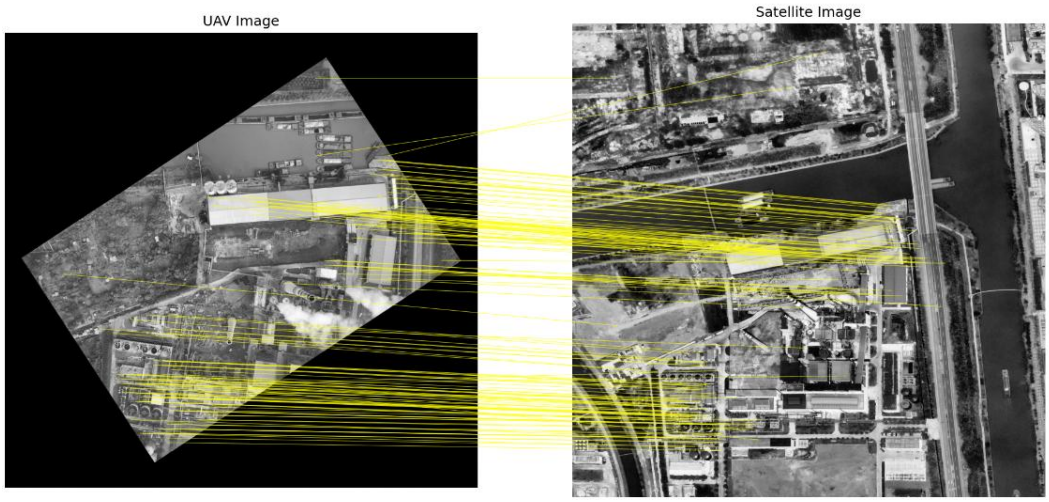
backbone. Scale factors for both horizontal and vertical axes are saved during this step to re-project coordinates back to the original image resolution later.

2. Model Setup and Execution: Pre-trained outdoor weights are loaded into the model to handle natural terrain features. The network is set to evaluation mode, transferred to available processing hardware (such as a GPU or CPU), and run in an inference-only context to maximize speed and efficiency.
3. Dense Correspondence Extraction: The preprocessed drone and satellite frames pass through the network as a paired input, generating a dense set of candidate pixel matches along with model-predicted confidence scores for each point pair.
4. Confidence Filtering and Coordinate Rescaling: Matches are filtered against a confidence threshold set at 50%. Unlike classical ratio tests that compare distance metrics between nearest neighbor descriptors, LoFTR confidence scores reflect predicted match probabilities refined to sub-pixel accuracy. Surviving coordinate pairs are then scaled back up using the saved spatial factors, placing them accurately on the original full-resolution images for downstream homography fitting.

Parameter	Value / Definition (config.py)	Purpose
LOFTR_MAX_IMAGE_SIZE	840 px	Upper bound on the longer image dimension fed to the network; balances match density/accuracy against memory footprint and inference latency.
LOFTR_CONFIDENCE_THRESHOLD	0.5	Minimum per-match confidence retained after inference; discards low-certainty correspondences prior to RANSAC, analogous in role to Lowe's ratio test.
pretrained weight set	'outdoor'	Kornia-hosted LoFTR checkpoint

Parameter	Value / Definition (config.py)	Purpose
		trained on outdoor scene pairs, better matched to the aerial/satellite imagery domain than the 'indoor' alternative.

Table 4.2 — LoFTR matching configuration.



4.4 Step 4: Spatial Homography and Outlier Rejection

Even after filtering matches based on model confidence, dense correspondence sets produced by deep learning matchers naturally retain a small fraction of false or geometrically inconsistent points. To compute a reliable planar homography matrix that accurately maps aerial coordinates onto the satellite reference frame, the system uses MAGSAC++, an advanced variation of the standard Random Sample Consensus (RANSAC) framework.

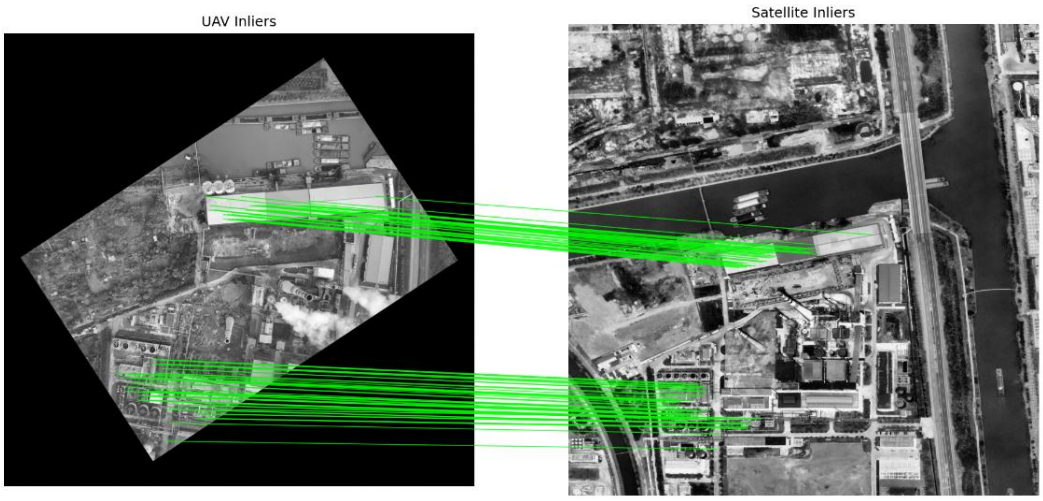
Unlike traditional RANSAC algorithms that rely on a fixed, hand-tuned distance threshold to label points as valid, MAGSAC++ evaluates matches across a continuous range of noise scales. This adaptive scoring approach makes it especially effective for deep-learning matchers, where spatial error patterns and false match distributions vary far more than those produced by classical keypoint detectors.

To ensure mathematical stability during matrix calculation, the system enforces a strict minimum match count before running the fitting algorithm. While four non-linearly aligned point matches represent the bare minimum needed to solve a projective transformation matrix, using a larger set of points builds an over-determined system that easily resists local spatial noise.

The fitting process returns the final homography matrix along with a binary indicator mask that isolates geometrically verified inlier points. This mask allows the system to compute precise inlier counts and accuracy percentages, while drawing a clean subset of surviving match lines across both images to provide a quick visual verification of structural alignment.

Parameter	Value / Definition (config.py)	Purpose
RANSAC_REPROJ_THRESHOLD	5.0 px	Reprojection error tolerance used by the MAGSAC++ scoring function when classifying a correspondence as inlier vs. outlier.
MIN_MATCH_COUNT	7	Minimum number of correspondences required before attempting homography estimation; guards against degenerate/under-determined fits.
Estimator	cv2.USAC_MAGSAC	Marginalizing sample consensus (MAGSAC++), avoiding a single fixed inlier threshold and improving robustness on dense, deep-learning-sourced correspondence sets.

Table 4.3 — RANSAC / homography estimation parameters (config.py).



4.5 Step 5: Warp and Footprint Projection

Once the system calculates the planar homography matrix, it projects the full-color drone frame directly into the spatial pixel grid of the reference satellite map. Executing this perspective transformation warps and aligns the aerial camera view onto the exact geometric plane of the satellite basemap.

To manage overlapping image boundaries cleanly, a matching binary mask of the original drone frame undergoes the exact same perspective transformation. This mask isolates the precise region occupied by the projected drone image, enabling an alpha-blending routine to blend the overlapping drone pixels with the underlying satellite map at a semi-transparent opacity level. The result is a seamless composite image that displays the aerial view layered directly over the satellite canvas.

In parallel, the system transforms the four corner points of the original drone frame using the calculated homography matrix. Connecting these four mapped points forms a closed bounding box on the satellite map, highlighting the drone's estimated ground footprint.

These visual outputs—the bounding footprint outline, the alpha-blended composite overlay, and the reference satellite map—act as a critical quality assurance check for the pipeline. A smooth, geometrically sound footprint outline gives clear visual proof that the transformation matrix was calculated accurately. Conversely, a heavily warped, stretched, or self-intersecting polygon immediately alerts engineers to upstream errors in feature matching or false-point filtering.



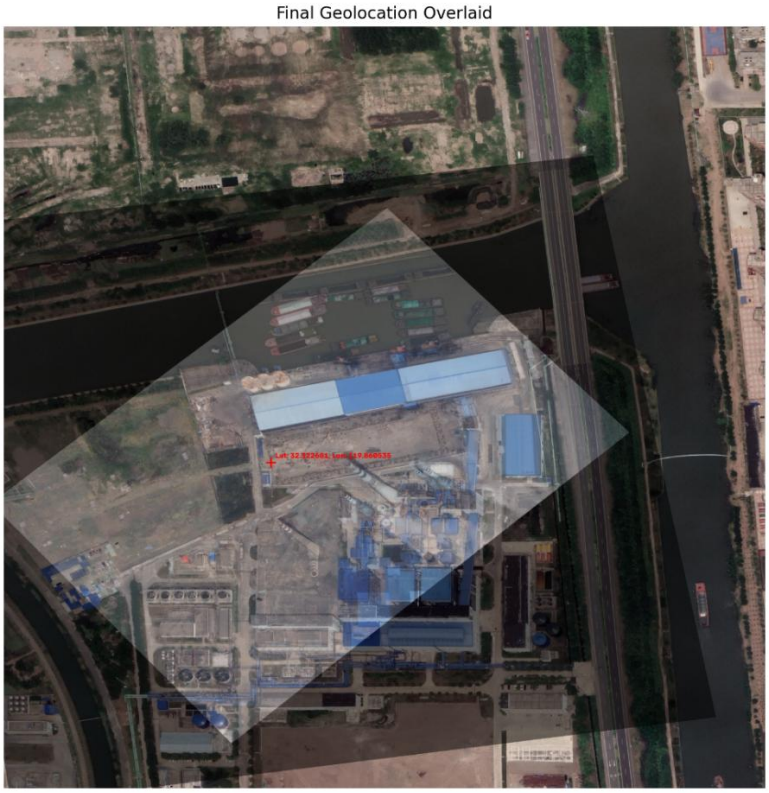
4.6 Step 6: Target Geolocation

The final phase translates a target pixel selected from the UAV frame into real-world geographic coordinates (latitude and longitude in the WGS84 system). Using the computed homography matrix, the target's pixel coordinates are projected onto the satellite base map's grid. The georeferencing engine then converts these satellite pixel coordinates into global geodetic coordinates using one of two modes:

- GeoTIFF Mode (Primary): When spatial metadata is present, the engine extracts the image's spatial metadata and affine transformation matrix. It translates pixel positions directly into coordinates and automatically reprojects them into standard WGS84 decimal degrees. This provides high geometric accuracy by relying on the map's original surveying calibration.

- Manual Calibration Mode (Fallback): If metadata is missing (such as in standard JPEG or PNG files), the system uses pre-configured geographic coordinates for the satellite image's four corners. It builds a localized perspective transformation matrix that maps any pixel coordinate within the frame to its corresponding latitude and longitude.

To confirm the output visually, a crosshair marker and coordinate label are drawn directly over the target location on the composite overlay image, producing an annotated map for final inspection.



1. Limitations & Error Analysis

5.1 System & Environmental Limitations

The georeferencing pipeline relies heavily on the **LoFTR** feature-matching framework, which imposes strict visual and structural constraints:

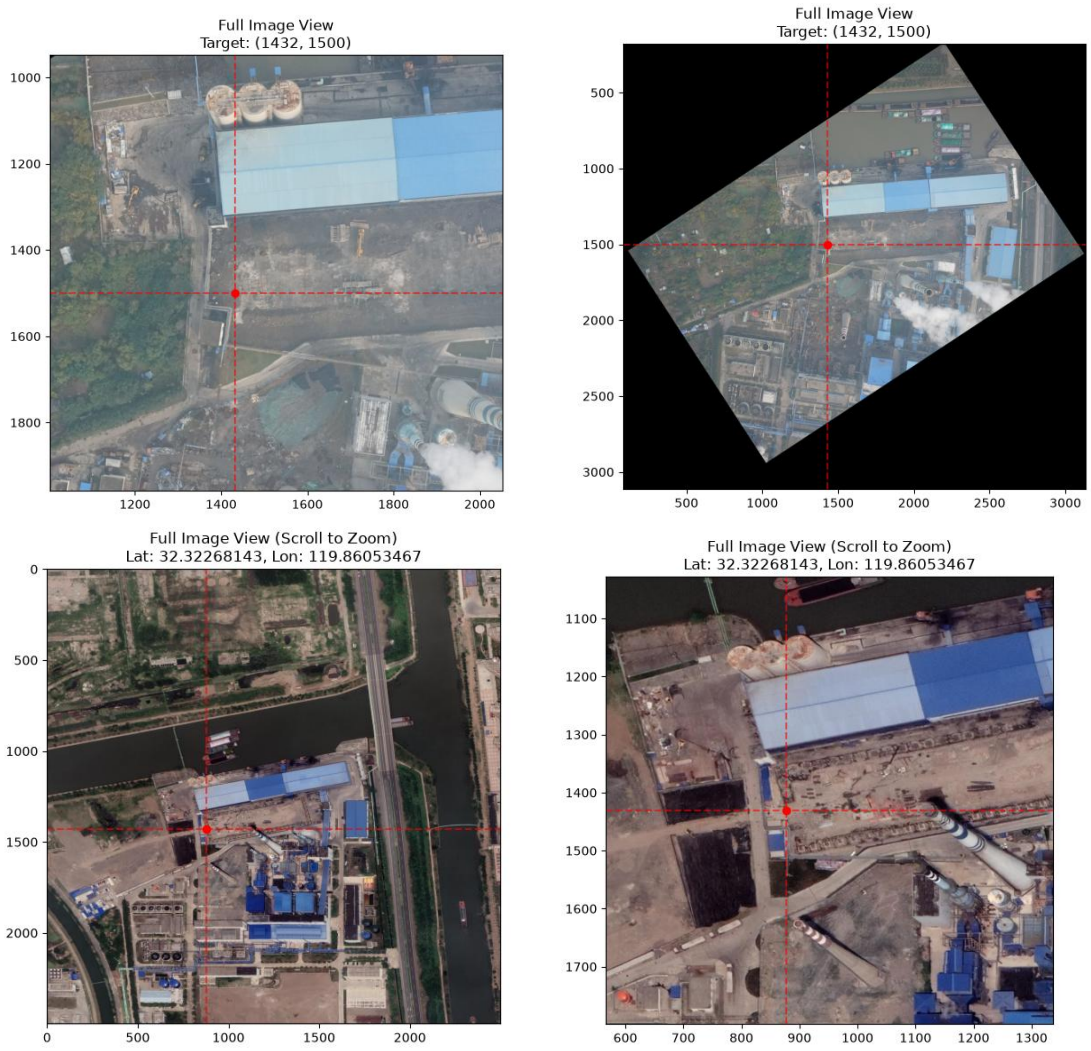
- Orientation & Perspective: Successful matching requires that the aerial (UAV) and satellite images share similar orientations. Furthermore, the UAV image must be captured from a near-nadir (top-down) view. Severe off-nadir angles or significant orientation mismatches prevent LoFTR from finding reliable feature correspondences.
- Environmental Distortions: Temporal gaps between captures introduce environmental challenges—such as seasonal variations, shifting shadow angles, extreme lighting differences, and structural modifications—that hinder feature alignment and degrade overall matching quality.

5.2 Error Analysis & Verification Protocol

Because intrinsic camera parameters and telemetry are omitted, geometric accuracy is evaluated using a three-way visual verification process:

1. UAV Input Pixel: Visually identifying and confirming the target pixel location on the raw aerial frame.
2. Calculated Geolocation: Computing the estimated latitude/longitude coordinates alongside their visually projected placement.
3. Satellite Basemap Reference: Inspecting the corresponding latitude/longitude coordinates directly on the reference satellite image.

Accuracy is confirmed by comparing all three representations—the raw UAV pixel, the calculated geographic output, and the true satellite location—to visually validate alignment correctness and isolate residual reprojection errors.



6. Conclusion & Future Work

6.1 Conclusion

This project presents a practical, end-to-end framework for geolocating raw UAV aerial images using reference satellite data.

The workflow begins with raw aerial imagery, which is preprocessed using on-board IMU data for orthorectification and rotational alignment. This step normalizes the camera angle to ensure the UAV frame closely matches the perspective of the reference satellite (.tif) map. Both images are then passed through our feature-matching and georeferencing pipeline to estimate the exact geographical coordinates of the target area.

Finally, the computed geolocation is verified through a mandatory visual check that cross-references the raw input pixel of the UAV image, its calculated latitude and longitude coordinates, and that exact geographic location on the satellite basemap.

Overall, combining IMU-assisted preprocessing with deep feature matching creates a reliable, fully visual positioning pipeline that works effectively without requiring active GPS.

6.2 Future Enhancements

- On-Board Real-Time Processing via Modified Sliding Window: Implement an optimized, modified sliding window technique to reduce latency and computational load. This will enable real-time processing directly on edge devices (on-board the UAV), making live georeferencing viable during flight.
- GPS-Denied SLAM-Based Autonomous Navigation: Integrate the visual georeferencing pipeline with a Simultaneous Localization and Mapping (SLAM) system. This will allow the aircraft to navigate autonomously in GPS-denied environments by constantly matching live camera feeds to satellite basemaps.
- Flight Telemetry & Metadata Integration: Incorporate embedded UAV sensor data—such as GPS coordinates, altitude, and tilt angles—to provide a initial location guess. This restricts the feature search area, dramatically speeding up match times and improving alignment stability.
- 3D Surface & Elevation Model (DEM) Integration: Incorporate digital elevation models to account for terrain variation, tall structures, and trees. Replacing the flat-earth assumption with 3D elevation data will eliminate visual displacement errors in non-flat landscapes.

Appendix A — Software Dependencies

Package	Constraint (requirements.txt)	Role in Pipeline
torch	>=1.10.0	Deep learning backend for LoFTR inference (CPU/CUDA tensor computation).
torchvision	(unpinned)	Companion library to torch; standard PyTorch vision utilities.
kornia	>=0.6.4	Provides kornia.feature.LoFTR and image/tensor geometry utilities used in Steps 2-3.
numpy	>=1.22	Array operations across all pipeline stages (points, masks, homogeneous coordinates).
opencv-contrib-python	>=4.7.0	CLAHE, perspective/homography estimation (incl. USAC_MAGSAC), warping, drawing.
matplotlib	>=3.5	All diagnostic visualizations saved at each pipeline stage.
rasterio	>=1.3	GeoTIFF affine transform / CRS reading and reprojection in geo_utils.py.

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[2] X. Qiu, S. Liao, D. Yang, Y. Li, and S. Wang, "Visual geo-localization and attitude estimation using satellite imagery and topographical elevation for unmanned aerial vehicles," *Eng. Appl. Artif. Intell.*, vol. 153, p. 110759, 2025, doi: 10.1016/j.engappai.2025.110759